

ATAKPAME, Togo

WMO: 653760

Lat: 7.583N

Lon: 1.117E

Elev: 1319

StdP: 14.01

Time Zone: 0.00 (E00)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	69.1	69.9	32.3	28.1	80.3	36.4	33.1	81.6	9.7	73.0	8.6	73.9	4.1	280	0.259

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.1	94.8	72.9	93.4	73.1	92.2	73.2	78.8	88.6	78.1	87.7	77.4	86.8	3.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.1	143.3	83.6	75.4	140.0	82.7	74.9	137.4	82.0	43.3	88.7	42.6	88.3	42.0	87.1	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	7.9	7.4	66.3	97.8	3.2	1.2	64.0	98.7	62.2	99.4	60.4	100.1	58.0	100.9		
(5)			50.6	81.9	2.7	1.9	48.6	83.3	47.0	84.4	45.5	85.5	43.5	86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.0	81.8	83.7	83.1	81.6	80.3	78.4	76.7	76.1	77.1	78.8	81.1	81.6
	DBStd	3.06	2.15	1.75	2.03	2.22	2.28	1.99	1.57	1.51	1.57	1.69	1.46	1.62
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10956	987	944	1028	949	940	852	828	809	814	892	932	981
	CDD65	5481	522	524	563	499	475	402	363	344	364	427	482	516
	CDH74	50079	5795	6273	6295	5149	4442	2977	2002	1793	2168	3233	4572	5380
(13)	CDH80	19143	2530	3106	2955	2111	1614	810	306	220	403	955	1871	2262
(14)	Wind	WSAvg	3.7	3.9	4.1	4.3	4.2	3.7	3.6	3.5	3.7	3.4	3.5	3.5
(15) Precipitation	PrecAvg	53.4	0.3	1.1	3.0	5.0	5.8	7.7	8.6	8.1	8.0	4.9	0.8	0.4
	PrecMax	81.4	2.4	4.7	7.6	11.2	12.8	13.3	17.1	14.4	15.0	11.4	3.9	4.5
	PrecMin	30.2	0.0	0.0	0.0	0.9	2.1	2.7	2.0	1.7	0.4	0.6	0.0	0.0
	PrecStd	11.1	0.5	1.2	2.0	2.2	2.2	2.7	3.8	3.4	3.5	2.6	0.9	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.2	96.8	96.8	93.6	92.0	89.2	86.2	85.5	86.9	89.6	92.6	93.4
		MCWB	70.1	72.1	73.9	76.9	77.5	77.3	75.8	75.4	76.0	76.7	74.6	71.5
	2%	DB	93.0	95.0	94.5	92.0	90.0	87.3	84.5	83.7	85.2	87.9	91.0	91.7
		MCWB	70.8	72.0	74.6	77.0	77.3	76.5	75.6	75.1	76.0	76.4	74.5	71.2
	5%	DB	91.3	93.6	93.1	90.5	88.8	86.1	83.3	82.4	83.9	86.6	89.8	90.4
		MCWB	70.5	71.8	74.6	77.0	77.0	76.2	75.3	74.8	75.8	76.2	74.2	70.6
	10%	DB	89.8	92.2	91.5	89.0	87.3	84.6	81.9	81.1	82.6	85.1	88.6	89.0
		MCWB	69.6	71.7	74.9	76.9	76.7	75.9	74.9	74.5	75.4	75.9	74.0	69.9
	0.4%	WB	76.5	78.0	79.1	80.1	79.5	78.3	77.4	77.1	77.9	78.2	77.9	77.2
		MCDB	88.6	90.9	91.2	89.1	88.2	86.6	83.8	82.8	83.7	86.8	88.3	88.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	75.4	76.8	77.8	78.9	78.4	77.4	76.5	76.1	76.9	77.4	76.9	75.9
		MCDB	86.8	89.0	89.6	88.8	87.5	85.4	82.7	81.7	83.4	85.7	87.5	86.9
	5%	WB	74.4	76.1	77.0	78.0	77.7	76.8	75.9	75.4	76.2	76.8	76.2	74.8
		MCDB	85.0	87.6	88.7	87.9	86.7	84.4	81.8	80.9	82.5	84.9	86.5	84.8
	10%	WB	73.6	75.3	76.3	77.3	77.1	76.2	75.2	74.8	75.5	76.2	75.5	73.9
		MCDB	83.3	86.0	87.4	86.9	85.8	83.3	81.0	79.9	81.6	83.9	85.4	82.7
	5% DB	MDBR	16.2	18.1	17.7	15.4	14.3	12.7	10.9	10.7	11.6	14.0	16.9	16.2
		MCDBR	19.0	20.2	19.7	17.2	16.2	14.6	13.1	12.9	13.5	15.9	18.1	18.3
		MCWBR	6.4	6.0	5.6	6.1	5.8	5.5	5.5	5.6	5.8	6.4	5.1	6.1
		MCWBR	18.6	18.6	17.9	16.0	15.2	13.8	11.9	11.6	12.7	14.9	16.9	17.5
(40) Clear-Sky Solar Irradiance	taub	0.755	0.778	0.806	0.647	0.545	0.501	0.479	0.473	0.468	0.518	0.574	0.643	
	taud	1.562	1.546	1.497	1.800	2.016	2.163	2.251	2.262	2.282	2.100	1.903	1.754	
	Ebn at Noon	195	197	194	225	243	251	258	264	269	255	237	218	
	Edh at Noon	88	92	97	71	56	47	44	44	44	52	62	72	
(44) All-Sky Solar Radiation	RadAvg	1620	1641	1716	1737	1666	1499	1338	1254	1398	1635	1679	1587	
(45)	RadStd	42	48	67	56	52	67	54	62	54	48	36	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.50	+0.56	N/A	+0.65	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page